If $f: \mathbb{R} \to \mathbb{R}$ satisfies Intermidiate Value Property and for any rational $r \in \mathbb{Q}$, $f^{-1}[\{r\}]$ is closed, then f is Continuous. (IVT property: If $f(a) < c < f(b)$, then $\exists x \in (a,b)$ s.t $c = f(x)$)

Proof. Suppose that f is not Continuous map. Then, there exists a sequence $\{x_n\}$ which Converges to x_0 but $f(x_n)$ not converges to $f(x_0)$. That is, there exists $\varepsilon > 0$ s.t

there exist infinitely many $n_i \in \mathbb{N}$ satisfying $|f(x_n) - f(x_0)| \geq \varepsilon$. $\left(\Leftrightarrow \begin{array}{c} f(x_0) \leq f(x_n) - \varepsilon < f(x_n) \\ \text{or} \\ f(x_0) \geq f(x_n) + \varepsilon > f(x_n) \end{array} \right)$

By the density of rationals, there exist two rationals $r^- \in (f(x_0) - \varepsilon, f(x_0))$ and $r^+ \in (f(x_0), f(x_0) + \varepsilon)$.

if $f(x_0) < f(x_{n_i})$, then $f(x_0) < r^+ < f(x_{n_i})$ and

if $f(x_{n_i}) < f(x_0)$, then $f(x_{n_i}) < r^- < f(x_0)$.

By the IVP, for each n_i, there exists $t_i \in (x_0, x_{n_i}) \cup (x_{n_i}, x_0)$ s.t $f(t_i) = r^+$ or r^-.

Since $x_{n_i} \to x_0$ as $i \to \infty$, $t_i \to x_0$ as $i \to \infty$.

For each $i \in \mathbb{N}$, $t_i \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}]$, closed set. therefore the limit point x_0 satisfies

$$x_0 \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}].$$

That is, $f(x_0) = r^+$ or r^-, it is Contradiction.

Note: Continuous preserves the Compactness, so, a continuous map $f: X \to \mathbb{R}$ on compact X has minimum and maximum, since $f[X]$ is closed bounded.

「Darboux's Theorem.

Let $f: [a,b] \to \mathbb{R}$ be a differentiable map.

Then, the derivative f' satisfies the Intermidiate Value Theorem.

Proof. Suppose that $f'(a) < f'(b)$. Let α s.t $f'(a) < \alpha < f'(b)$ be given.

Define $\quad g(x) = f(x) - \alpha x \quad (x \in [a,b])$.

Clearly g is differentiable, and $g'(a) = f'(a) - \alpha < 0$, $g'(b) = f'(b) - \alpha > 0$.

And, since g is continuous, it has a minimum in $[a,b]$, for some c.

By the assumption, $c \neq a$ and $c \neq b$. And, since g has a minimum at c,

$g'(c) = f'(c) - \alpha = 0$. So proved $\boxtimes$

「Corollary. If $f: I \to \mathbb{R}$ has a primitive, then $f[I]$ is an interval.

Note: If f is defined on an interval and cannot satisfy IVT,
then it has no primitive.